

**SIMATS ENGINEERING (SAVEETHA SCHOOL OF
ENGINEERING)** Department of Computer Science and Engineering

ASSESSMENT TOOL 1 – ARCHITECTURE / MODELING (INFRASTRUCTURE DESIGN)

Course Code:	CSA5802 Course Title: DEVSECOPS ENGINEERING AND APPLICATION SECURITY
CO Assessed:	CO4 – Precision (BL1, BL2, BL3) Assessment: Assessment Tool 1
Weightage:	30% of CIA Total Marks: 100
CO4 Statement:	Design a secure infrastructure architecture for a College Campus Network showing student, faculty, lab, server, guest Wi-Fi, -Fi. and and administration VLANs with h firewalls. NAC. IDS/IPS. and centralized logging using Draw.io/Diagrammer
Student Name:	K.Yalla Reddy Reg. No.: 192465047
Section / Batch:	2024 Date: 19-08-2026

Instructions to Students

- Create an original secure infrastructure architecture using DiagrammeR or Draw.io.
- Clearly represent major components, network zones, trust boundaries, data flows and security controls.
- Include IAM, encryption, monitoring/logging and secure communication paths wherever applicable.
- Use clear labels, consistent symbols and professional alignment so the architecture is easy to interpret.
- Submit the editable diagram source file along with the final exported PDF through the prescribed submission platform.

ARCHITECTURE /

MODELING Architecture Design Q1 Q1 – 100

RUBRICS FOR CALCULATION

Criteria	Wt.	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Infrastructure Architecture Design	20%	Designs a comprehensive, logically structured and technically accurate secure infrastructure architecture with all essential components	Designs a clear and technically sound infrastructure architecture	Designs a basic infrastructure architecture with limited	Provides an incomplete, unclear or technically inaccurate infrastructure architecture

			with only components or minor noticeable structural gaps component or structural omissions	
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Security Controls and Network Segmentation	30%	Correctly integrates layered security controls, network segmentation, trust boundaries and secure communication paths throughout the architecture	Integrates appropriate security controls and segmentation with only minor gaps	Includes basic security controls but shows limited segmentation or incomplete trust boundary representation	Includes insufficient or inappropriate security controls with poor network segmentation
IAM, Data Protection and Monitoring	20%	Accurately represents IAM, encryption, secure data storage, logging and monitoring mechanisms with clear component relationships	Represents most IAM, data protection and monitoring mechanisms correctly with minor omissions	Represents basic IAM, data protection or monitoring elements but several important mechanisms are missing or unclear	Provides limited or incorrect representation of IAM, data protection and monitoring mechanisms
Diagram Quality and Tool Utilization	20%	Uses DiagrammeR or Draw.io effectively to produce a professional, readable and well organized architecture diagram with clear labels and connections	Produces a clear and organized diagram with effective tool usage and only minor presentation issues	Produces a basic diagram with inconsistent layout, labeling or tool utilization	Produces a poorly organized or difficult-to-understand diagram with inadequate tool utilization

Documentation and Technical Justification	10%	Clearly documents architecture components and provides strong technical justification for security design decisions	Provides clear documentation and appropriate justification with minor gaps	Provides basic documentation with limited technical justification	Provides incomplete or unclear documentation with little or no technical justification
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Rubrics:

Criteria Weightage Q5 (10) Total Marks (50)	
Infrastructure Architecture Design 25%	
Security Controls and Network Segmentation	30%
IAM, Data Protection and Monitoring	20%
Diagram Quality and Tool Utilization	15%
Documentation and Technical Justification	10%
Total per Question 100 Marks	

100

CSA5802 - DEVSECOPS ENGINEERING AND APPLICATION SECURITY

CO3- ASSESSMENT TASK 1:

DESIGN A SECURE INFRASTRUCTURE ARCHITECTURE FOR A COLLEGE CAMPUS NETWORK SHOWING STUDENT, FACULTY, LAB, SERVER, GUEST WI-FI, -FI. AND AND ADMINISTRATION VLANS WITH H FIREWALLS. NAC. IDS/IPS. AND CENTRALIZED LOGGING USING

DRAW.IO/DIAGRAMMER

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ABSTRACT

A modern college campus network contains a large number of users, devices, applications and services. Students, faculty members, administrators, laboratory systems and guests require different levels of access to network resources. If all these users and devices operate on a single flat network, a compromised device can potentially provide an attacker with access to sensitive systems.

This report proposes a secure infrastructure architecture for a college campus network using network segmentation, Virtual Local Area Networks (VLANs), Next-Generation Firewalls (NGFW), Network Access Control (NAC), Intrusion Detection and Prevention Systems (IDS/IPS), Identity and Access Management (IAM), encryption and centralized security monitoring. The proposed architecture separates student, faculty, laboratory, administration, server, guest and security-management networks into different security zones.

The architecture follows a defense-in-depth approach. A perimeter firewall protects the campus from external threats, while an internal segmentation firewall protects critical resources. NAC verifies devices before network access is provided, and IAM controls user authentication and authorization. IDS/IPS monitors network traffic for suspicious activity, while a centralized SIEM platform collects security logs from network and security components.

The proposed architecture improves confidentiality, integrity and availability while reducing the attack surface of the campus network. The architecture is designed using Draw.io and provides clear trust boundaries, data flows and security controls.

1. INTRODUCTION

A college campus network is a complex environment consisting of students, faculty, administrative employees, laboratories, servers, wireless devices, Internet services and network infrastructure. These components exchange large amounts of information every day.

Important information handled by a college may include student records, examination information, financial information, employee information, academic records, research data and authentication credentials. Therefore, protecting the campus network is an important cybersecurity requirement.

Traditional flat networks provide limited isolation between different groups of users. For example, if a student's laptop becomes infected with malware, the attacker may attempt to scan and attack other systems on the same network. Similarly, guest users should not have access to internal college systems.

The proposed architecture addresses these problems through network segmentation and multiple layers of security. The architecture separates the network into dedicated VLANs for students, faculty, laboratories, administration, servers, guests and security management. Firewalls control communication between zones, NAC verifies devices, IAM manages identities and access privileges, IDS/IPS detects malicious traffic, and SIEM provides centralized monitoring.

The architecture therefore follows the principle of **defense in depth**, where multiple security mechanisms work together instead of relying on a single security control.

2. OBJECTIVES

The main objectives of the proposed architecture are:

1. To design a secure infrastructure architecture for a college campus network.
2. To separate student, faculty, laboratory, administration and guest traffic using VLANs.
3. To protect critical servers using firewall-based network segmentation.
4. To implement Network Access Control for authorized devices.
5. To implement IAM with authentication, MFA and role-based access control.
6. To detect and prevent malicious network activity using IDS/IPS.
7. To provide centralized security logging and monitoring using SIEM.
8. To protect sensitive data using encryption.
9. To establish clear trust boundaries between network zones.
10. To provide secure communication between users and applications.
11. To reduce the attack surface of the campus network.
12. To provide a scalable architecture that can be expanded in the future.

3. EXISTING CAMPUS NETWORK SECURITY CHALLENGES

A campus network can face several security challenges.

3.1 Flat Network Architecture

If all systems use the same network, there is little isolation between users. A compromised student device could potentially communicate with administrative or server systems.

3.2 Unauthorized Device Access

Students or visitors may connect unauthorized laptops, mobile devices or other systems to campus network ports.

3.3 Malware and Ransomware

Malware can enter through infected devices, malicious websites, email attachments or compromised applications.

3.4 Credential Theft

Attackers may attempt phishing, password attacks or credential theft to obtain user

accounts.

3.5 Guest Network Risks

Guest users require Internet connectivity but should not be able to access internal college resources.

3.6 Insider Threats

A legitimate user may attempt to access resources outside their authorized role.

3.7 Lack of Centralized Monitoring

If security logs are stored independently on different devices, detecting a coordinated attack becomes difficult.

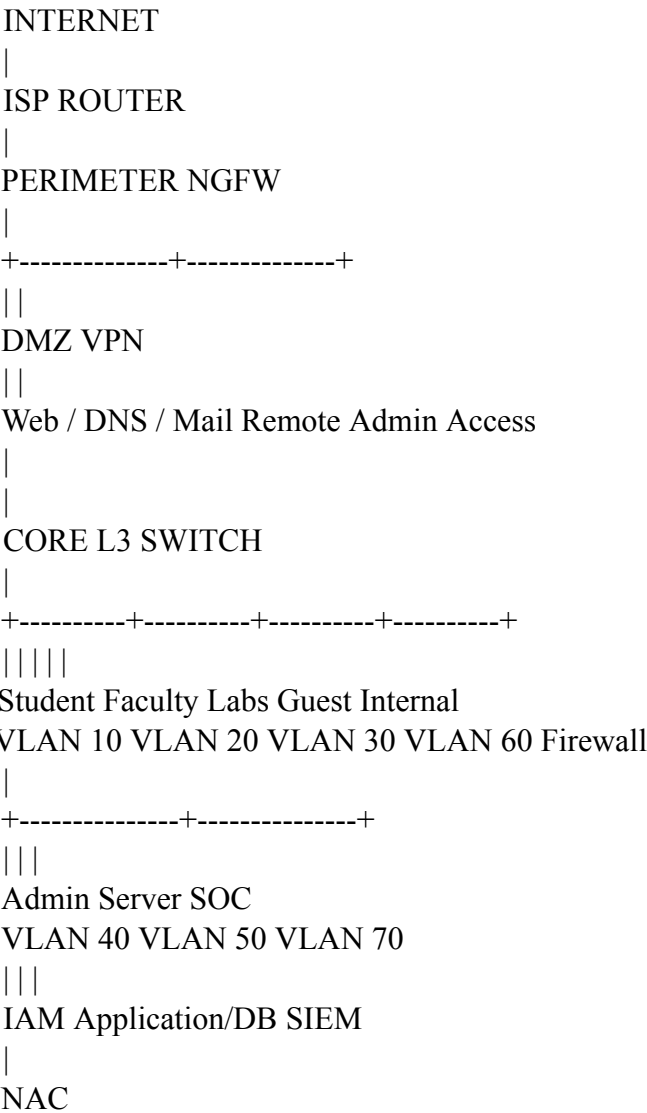
3.8 Sensitive Data Exposure

Student records, financial data, examination information and research data require appropriate access controls and encryption.

4. PROPOSED SECURE CAMPUS NETWORK ARCHITECTURE

The proposed architecture contains several security zones.

High-Level Architecture



The architecture is divided into zones to limit unauthorized communication.

5. NETWORK SEGMENTATION AND VLAN DESIGN

Network segmentation is one of the most important security controls in the proposed architecture. Each major user group is assigned a dedicated VLAN.

VLAN Network Example Subnet Purpose

VLAN 10 Student 10.10.10.0/24 Student devices

VLAN 20 Faculty 10.10.20.0/24 Faculty devices

VLAN 30 Laboratory 10.10.30.0/24 Computer laboratory systems VLAN 40

Administration 10.10.40.0/24 Administrative systems VLAN 50 Servers

10.10.50.0/24 Critical servers

VLAN 60 Guest 10.10.60.0/24 Guest Wi-Fi

VLAN 70 Security Management 10.10.70.0/24 SIEM and security tools VLAN 80

Network Management 10.10.80.0/24 Network infrastructure management

5.1 Student VLAN

The Student VLAN contains student laptops, desktops and authorized wireless devices.

Students can access:

- Internet
- Learning Management System
- Digital library
- Approved academic applications

Students cannot directly access:

- Administration systems
- Security management systems
- Network management systems
- Sensitive databases

5.2 Faculty VLAN

The Faculty VLAN contains faculty workstations and authorized

devices. Faculty users can access:

- LMS
- Department applications
- Academic file servers
- Approved Internet services

Faculty access is controlled using IAM, MFA and RBAC.

5.3 Laboratory VLAN

The Laboratory VLAN contains computers used by students for practical sessions.

The laboratory network is isolated from administration and critical server networks.

Security controls include:

- NAC
- Endpoint security
- Web filtering
- Firewall policies
- IDS/IPS monitoring

5.4 Administration VLAN

The Administration VLAN is a high-trust network containing:

- Finance systems
- HR systems
- Student records
- Examination systems
- Administrative applications

Access to this network requires stronger security controls including MFA, RBAC and firewall policies.

5.5 Server VLAN

The Server VLAN contains critical infrastructure such as:

- Application servers
- LMS servers
- Database servers
- File servers
- Authentication servers
- Backup servers
- DNS/DHCP servers

The Server VLAN is protected using an internal segmentation firewall.

5.6 Guest VLAN

The Guest VLAN provides Internet access to visitors.

Guest traffic is isolated from all internal college VLANs.

The primary policy is:

Guest → Internet: Allowed

Guest → Internal Network: Denied

6. SECURITY CONTROLS

The proposed architecture uses multiple security mechanisms.

6.1 Next-Generation Firewall

The perimeter NGFW provides:

- Stateful packet inspection
- Application filtering
- Access control
- URL filtering
- VPN
- Threat prevention
- Network address translation
- Traffic logging

The firewall is positioned between the Internet and internal campus network.

6.2 Internal Segmentation Firewall

An internal firewall is placed between the general campus network and critical administration/server zones.

Its purpose is to control traffic between trusted and less-trusted network segments. For example:

Student VLAN

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X

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Administration VLAN

Direct student-to-administration traffic is blocked.

6.3 Network Access Control

NAC ensures that only authorized and compliant devices connect to the campus network.

The process is:

Device

↓

NAC

↓

Identity Verification

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Device Compliance Check

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VLAN Assignment

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Network Access

An unauthorized device can be denied or placed into a quarantine VLAN.

6.4 IDS/IPS

IDS/IPS provides network threat detection and prevention. It can identify:

- Port scanning
- Brute-force attempts
- Malware traffic
- Suspicious connections
- Denial-of-service patterns
- Known attack signatures

IDS generates alerts, while IPS can actively block suspicious traffic.

7. IDENTITY AND ACCESS MANAGEMENT

IAM provides centralized control over users and their permissions.

The architecture uses:

Authentication

Users authenticate using their college credentials.

Multi-Factor Authentication

MFA provides an additional security factor beyond a password.

Role-Based Access Control

Access permissions are assigned based on user roles.

Example:

User Role Access

Student LMS, Library, Internet

Faculty LMS, Academic Systems

Administrator Administrative Applications

Network Administrator Network Management

Security Analyst SIEM and Security Logs

Guest Internet Only

This follows the principle of **least privilege**, where users receive only the access necessary for their responsibilities.

8. DATA PROTECTION AND ENCRYPTION

Sensitive information should be protected both during transmission and while stored.

8.1 Data in Transit

HTTPS and TLS are used for communication between users and applications.

Example:

Student Device

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| HTTPS/TLS

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LMS Application

8.2 Data at Rest

Sensitive databases and backup data should be encrypted.

Examples include:

- Student records
- Financial information
- Examination records
- Employee information
- Research data

8.3 Backup Protection

Backup servers should be isolated and backups should be encrypted.

A secure backup architecture reduces the impact of ransomware and accidental data loss.

9. CENTRALIZED LOGGING AND MONITORING

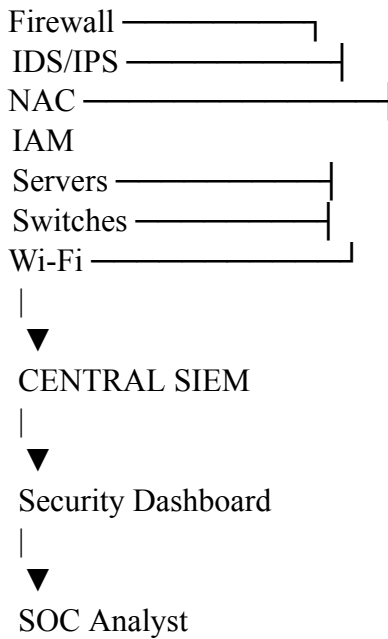
Centralized monitoring is implemented using a SIEM platform.

Security logs are collected from:

- Firewalls
- IDS/IPS
- NAC
- IAM
- Servers

- Switches
- Wireless access points
- Database systems

Architecture:



Secure logging protocols should be used to protect log transmission. The SIEM can identify patterns such as:

- Repeated failed logins
- Unusual network connections
- Multiple IDS alerts
- Suspicious administrative activity
- Abnormal traffic patterns

10. SECURE COMMUNICATION AND DATA FLOW

The architecture uses controlled communication paths.

Student to LMS

Student VLAN

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Firewall / ACL

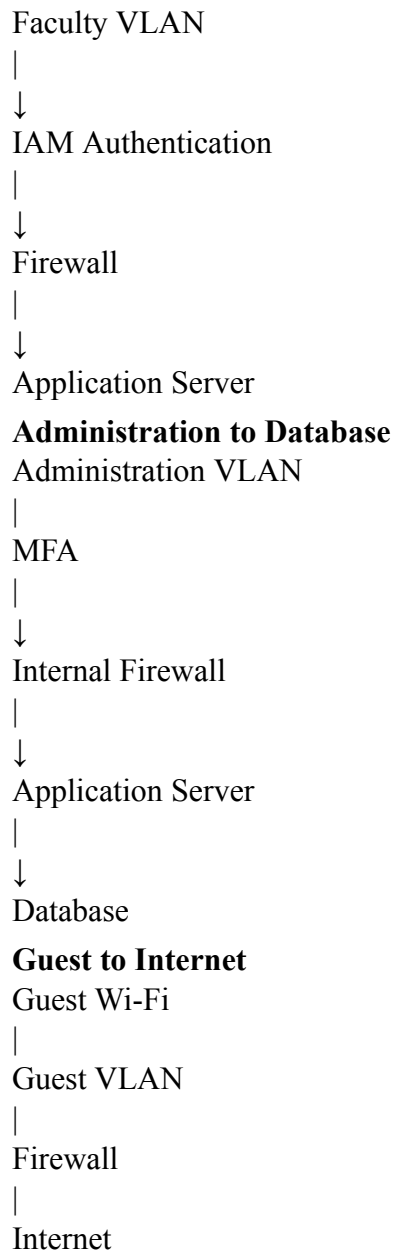
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LMS Server

Communication uses HTTPS/TLS.

Faculty to Academic Server



The guest network has no direct route to internal systems. **11. TRUST BOUNDARIES**

Trust boundaries indicate where different levels of trust exist within the architecture.

Boundary 1 – Internet to Campus

The perimeter firewall separates the untrusted Internet from the college

network. **Boundary 2 – DMZ to Internal Network**

Public-facing systems are separated from internal systems.

Boundary 3 – User VLANs to Critical VLANs

Student, faculty and laboratory systems are separated from administration and server systems. **Boundary 4 – General Network to Security Network** Security

monitoring systems are placed in a restricted management zone. **Boundary 5 –**

Guest to Internal Network

Guest Wi-Fi is completely isolated from internal college resources.

These boundaries reduce lateral movement during a security incident.

12. THREAT ANALYSIS AND MITIGATION

Threat	Possible Impact	Security Control	Malware	System compromise
Endpoint security,	IDS/IPS	Phishing	Credential theft	IAM, MFA, email
security	Brute-force attack	Account compromise	MFA, rate limiting,	
SIEM	Port scanning	Network reconnaissance	Firewall, IDS/IPS	
Unauthorized device	Network compromise	NAC		

Insider threat	Data exposure	RBAC, logging, least privilege	Ransomware	Data loss
Segmentation, backups,	IDS/IPS	Guest attack	Internal compromise	Guest
VLAN isolation	Database attack	Sensitive data theft	Firewall, encryption,	
RBAC	DDoS	Service disruption	NGFW, traffic filtering	Credential theft
Unauthorized access	MFA and IAM			

13. ARCHITECTURE WORKFLOW

The overall security workflow is:

Step 1 – Device Connection

A user connects a laptop or other device to the campus network. **Step 2 – NAC Verification**

NAC verifies the device and user.

Step 3 – Authentication

IAM authenticates the user's identity.

Step 4 – Role Verification

RBAC determines which resources the user is permitted to

access. **Step 5 – VLAN Assignment**

The user is placed into the appropriate VLAN.

Step 6 – Firewall Inspection

Traffic is evaluated against firewall policies and ACLs.

Step 7 – IDS/IPS Inspection

Network traffic is monitored for malicious activity.

Step 8 – Application Access

The user accesses authorized applications.

Step 9 – Logging

Relevant security events are forwarded to the SIEM.

Step 10 – Alert and Response

If suspicious behavior is detected, the SIEM generates an alert for the security team.

14. IMPLEMENTATION PLAN

The architecture can be implemented in phases.

Phase 1 – Network Planning

- Identify users and devices.
- Identify critical applications.
- Identify security requirements.
- Define VLANs and IP address ranges.

Phase 2 – VLAN Implementation

Create separate VLANs for:

- Students
- Faculty
- Labs
- Administration
- Servers
- Guests
- Security management
- Network management

Phase 3 – Firewall Configuration

Configure:

- Internet firewall policies
- Inter-VLAN ACLs
- DMZ rules
- Server access rules
- Guest isolation

Phase 4 – NAC Deployment

Deploy NAC and configure:

- 802.1X
- Device authentication
- Device compliance
- Quarantine policies

Phase 5 – IAM Deployment

Implement:

- Central authentication
- MFA
- RBAC
- Administrative access controls

Phase 6 – IDS/IPS Deployment

Deploy monitoring sensors at:

- Internet perimeter
- Internal segmentation points
- Server network

Phase 7 – SIEM Deployment

Configure centralized collection of security logs.

Phase 8 – Testing

Conduct:

- VLAN isolation testing
- Firewall testing
- Unauthorized device testing
- Guest isolation testing
- Authentication testing
- IDS/IPS testing
- Logging verification

15. TECHNICAL JUSTIFICATION

The proposed architecture follows a defense-in-depth security strategy.

VLAN segmentation prevents unnecessary communication between different user groups. The perimeter firewall protects the network from external threats, while the internal firewall provides additional protection for critical administration and server systems.

NAC prevents unauthorized devices from gaining unrestricted network access. IAM provides centralized authentication and authorization, while MFA reduces the risk associated with stolen passwords.

IDS/IPS provides continuous inspection of network traffic and helps identify malicious activity. Centralized SIEM monitoring allows security personnel to correlate events from multiple devices and investigate potential incidents. Encryption protects sensitive information during transmission and storage. Guest isolation prevents visitors from accessing internal systems.

The architecture therefore combines preventive, detective and corrective security mechanisms.

16. ADVANTAGES OF THE PROPOSED ARCHITECTURE

The proposed architecture provides the following advantages:

16.1 Improved Security

Multiple security controls reduce the probability of successful attacks.

16.2 Network Isolation

VLAN segmentation prevents unnecessary communication between departments.

16.3 Reduced Lateral Movement

If one device becomes compromised, segmentation makes it more difficult for attackers to move toward critical systems.

16.4 Better Access Control

IAM, MFA, RBAC and NAC ensure that only authorized users and devices receive appropriate access.

16.5 Improved Monitoring

Centralized SIEM monitoring provides a unified view of security events.

16.6 Guest Protection

Guest users can access the Internet without accessing internal college resources.

16.7 Protection of Critical Data

Administration and server networks receive stronger security controls.

16.8 Scalability

Additional departments, VLANs, servers and security controls can be added as the college network grows.

17. LIMITATIONS

Although the proposed architecture provides strong security, several limitations should be considered.

1. Implementation requires investment in security hardware and software.
2. Network configuration can become complex as the number of VLANs increases.
3. NAC and IAM require proper configuration and maintenance.
4. SIEM systems can generate a large number of alerts and require skilled security personnel.
5. Firewall and IDS/IPS rules must be regularly updated.
6. Employees and students require cybersecurity awareness training.
7. Security controls can affect network performance if incorrectly configured.

18. FUTURE ENHANCEMENTS

The architecture can be further improved by implementing:

- Zero Trust Network Access
- Security Orchestration, Automation and Response (SOAR)
- Endpoint Detection and Response (EDR)
- Cloud security controls
- Vulnerability management
- Automated incident response
- Network behavior analytics
- Advanced threat intelligence
- Automated compliance monitoring
- Micro-segmentation
- Continuous security assessment

A Zero Trust approach can further strengthen the architecture by continuously verifying users, devices and access requests instead of automatically trusting users based only on their network location.

19. CONCLUSION

This report presented a secure infrastructure architecture for a college campus network. The architecture separates students, faculty, laboratories, administration, servers, guests and security-management systems into dedicated network zones using VLAN segmentation.

A layered security approach is implemented using Next-Generation Firewalls, internal segmentation firewalls, Network Access Control, Identity and Access Management, Multi-Factor Authentication, Role-Based Access Control, IDS/IPS, encryption and centralized SIEM monitoring.

The architecture provides clear trust boundaries and controlled data flows between different network zones. Sensitive administration and server systems receive stronger protection, while guest users are isolated from internal resources.

The proposed architecture improves confidentiality by restricting access to sensitive information, integrity through authentication and monitoring, and availability through controlled network design and secure backup mechanisms.

Overall, the architecture demonstrates how security can be incorporated into the infrastructure design rather than being added only after deployment. The design therefore provides a practical foundation for a secure, scalable and manageable college campus network.

20. REFERENCES

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